CHATFIELD RESEARCH

Week 3

CHATFIELD Quantitative Research

Division Updates & Research Project Launch



PROJECT UPDATES



RESEARCH PROJECT LAUNCH



TEAM DEVELOPMENT



KEY FOCUS AREAS



NEXT STEPS

21 October 2026

SECTOR ARTICLE ASSIGNMENTS

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Each analyst will take responsibility for one of Chatfield's four research sectors



MACRO ECONOMICS

Research into the wider economy and financial environment.

- Growth
- Inflation
- Interest rates
- Employment
- Monetary policy
- Fiscal policy



FIXED INCOME & CREDIT

Research into debt markets and the factors affecting their pricing and risk.

- Government bonds
- Corporate bonds
- Yield curves
- Interest rates
- Credit risk
- Credit spreads



COMMODITIES & ENERGY

Research into physical commodity and energy markets.

- Oil
- Gas
- Metals
- Agriculture
- Supply & demand
- Production
- Geopolitics



STUDENT ECONOMICS

Research into economic and financial issues directly affecting students.

- Cost of living
- Housing & rent
- Employment
- Student finance
- Spending

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EXPANSION OF THE QUANTITATIVE RESEARCH DIVISION

Building a larger and more impactful Quant division for 2026/27

CURRENT TEAM

- Head of Quantitative Research
  - James Davies (MSci, final year)
- Four Senior Quant Analysts
  - One analyst per research sector
  - Third-year students
  - Responsible for sector analysis and wider research output

EXPANSION PLANS

- Recruit additional Quantitative Research members across 2026/27
- Open to undergraduate students with strong quantitative skills and interest in financial research
- Support analysts within each sector and on collaborative projects
- Provide opportunities for skill development and research experience

KEY OBJECTIVES

- Increase the size and capability of the Quant division
- Maintain high-quality, rigorous research output
- Develop a pipeline of quantitative talent within the society
- Create opportunities for leadership, collaboration and technical skill development

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OUR FIRST COLLABORATIVE RESEARCH PROJECT

Implementing and improving an existing time-series forecasting model

MOTIVATION

Our first project provides a practical, achievable and valuable opportunity to:

- Develop applied quantitative research skills
- Work collaboratively across sectors
- Produce a publishable article for Chatfield Research
- Build towards more advanced research in second term

PROJECT OVERVIEW

Each analyst will implement an existing forecasting model from an academic paper, applied to a chosen economic dataset within their sector.

We will aim to replicate the original results and make modest, well-justified improvements.

CANDIDATE MODELS

- Dynamic Nelson-Siegel yield curve model
- Bayesian structural time series (BSTS)
- ARIMA/VAR with exogenous indicators

EXPECTED OUTPUT

A high-quality research article, including:

- Clear economic motivation
- Data description and methodology
- Model implementation and results
- Evaluation and comparison to benchmark
- Discussion of findings and limitations

SUCCESS CRITERIA

1. Replicate the core results from the paper
2. Provide a clear and well-structured analysis
3. Achieve a meaningful improvement or extension

RESEARCH METHODOLOGY: ARIMA vs ARIMAX

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Two time-series models to forecast key economic indicators



ARIMA
(Autoregressive Integrated Moving Average)

A univariate time-series model that uses only past values of the target variable to make forecasts.

$$\phi(B)(1-B)^d y_t = \theta(B)\varepsilon_t$$

- Captures autocorrelation in the data
- Suitable for stationary (or differenced) series
- Defined by three parameters (p, d, q)
- Strong benchmark and widely used in economic forecasting



ARIMAX
(ARIMA with Exogenous Variables)

An extension of ARIMA that includes external (exogenous) variables to improve forecast accuracy.

$$\phi(B)(1-B)^d y_t = \theta(B)\varepsilon_t + \beta' X_t$$

- Includes external explanatory variables
- Can capture the impact of economic, financial or sector-specific indicators
- Often improves forecast accuracy
- Useful when relevant external data is available



KEY DIFFERENCES

Feature	ARIMA	ARIMAX
Input variables	Past values only	Past values + exogenous variables
Model type	Univariate	Multivariate (with exogenous inputs)
Use case	Baseline model	When relevant external data is available
Complexity	Lower	Higher
Forecast potential	Good	Often better (when exogenous variables are informative)

OUR APPROACH

1. Implement and evaluate both ARIMA and ARIMAX for the chosen economic dataset.
2. Compare forecast performance using out-of-sample testing and standard evaluation metrics.

SIX-WEEK RESEARCH TIMELINE

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From implementation to final article

WEEK 1
21 – 27 OCT

LITERATURE REVIEW & DATA COLLECTION

- Read and understand the chosen paper
- Collect and explore economic dataset
- Reproduce baseline model structure
- Plan implementation details

WEEK 2
28 OCT – 3 NOV

DATA PREPARATION & INITIAL MODEL

- Clean and preprocess data
- Implement baseline ARIMA model
- Run initial results and diagnostics
- Document methodology and assumptions

WEEK 3
4 – 10 NOV

MODEL EXTENSION (ARIMAX)

- Incorporate relevant exogenous variables
- Implement ARIMAX model
- Compare to baseline ARIMA results
- Initial analysis of forecast performance

WEEK 4
11 – 17 NOV

RESULTS & ANALYSIS

- Full model estimation and diagnostics
- Out-of-sample forecast evaluation
- Compare ARIMA vs ARIMAX performance
- Analyse key economic insights and implications

WEEK 5
18 – 24 NOV

ARTICLE WRITING

- Write up methodology, results and discussion
- Create figures and tables
- Link to wider economic context
- Prepare draft article for review

WEEK 6
25 NOV – 1 DEC

FINALISATION & PUBLICATION

- Incorporate feedback and refinements
- Finalise article and visualisations
- Prepare presentation for Open Session
- Submit article for publication

KEY MILESTONES

- **Week 1** Literature review complete
- **Week 2** Baseline model implemented
- **Week 3** ARIMAX model implemented
- **Week 4** Results and comparison complete
- **Week 5** Draft article ready
- **Week 6** Final article and presentation

THREE INDEPENDENT RESEARCH ROLES

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Each analyst will focus on a specific aspect of the project in Week 1, working together to produce a complete research article



QUANT 1
Economic Data & Time-Series Analysis

Responsible for sourcing, cleaning and exploring the economic dataset.

KEY RESPONSIBILITIES

- Identify and collect relevant economic data
- Clean and preprocess the dataset
- Conduct initial exploratory data analysis
- Analyse key time-series properties (stationarity, seasonality, trends)
- Prepare dataset for modelling



QUANT 2
Statistical Modelling & Methodology

Responsible for implementing and comparing the ARIMA and ARIMAX models.

KEY RESPONSIBILITIES

- Implement the baseline ARIMA model
- Incorporate exogenous variables to build the ARIMAX model
- Estimate model parameters and run diagnostics
- Compare model performance using out-of-sample testing
- Justify model choices and methodology



QUANT 3
Python Implementation & Forecast Evaluation

Responsible for translating the models into Python and evaluating the forecasts.

KEY RESPONSIBILITIES

- Implement ARIMA and ARIMAX models in Python
- Run forecasting and generate out-of-sample predictions
- Evaluate forecast performance using standard metrics (e.g. RMSE, MAE, MAPE)
- Produce clear visualisations of results
- Support integration of all components into the final article

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WEEK 1 – QUANT 1: ECONOMIC DATA & TIME-SERIES ANALYSIS

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Data collection, cleaning and initial exploratory analysis

OBJECTIVE

Source, clean and explore the chosen economic dataset to understand its key features and time-series properties.

KEY TASKS

1. Identify and collect the chosen economic dataset for our project
2. Clean and preprocess the data (handle missing values, outliers, transformations)
3. Produce descriptive statistics and visualisations
4. Analyse key time-series properties (stationarity, trend, seasonality, volatility)
5. Prepare a clean, well-documented dataset for modelling

KEY DATA SOURCES (OPTIONS)

- **FRED – Macroeconomic Data (US)**
  https://fred.stlouisfed.org/
- **BEA – US Economic Indicators**
  https://www.bea.gov/data
- **Eurostat – European Economic Data**
  https://ec.europa.eu/eurostat/
- **ONS – UK Economic Data**
  https://www.ons.gov.uk/
- **World Bank – Global Indicators**
  https://data.worldbank.org/

EXAMPLE ECONOMIC AREAS

- Macroeconomic indicators (e.g. GDP, inflation)
- Financial markets (e.g. interest rates, equity indices)
- Energy & commodities (e.g. oil, gas prices)

EXPECTED DELIVERABLES

- Clean, well-structured dataset (CSV/Parquet) with clear documentation
- Exploratory data analysis report/notebook (including key visualisations)
- Summary of key time-series properties and initial observations
- Brief discussion of data limitations and any preprocessing decisions

USEFUL RESOURCES

- Pandas documentation (data cleaning)
  https://pandas.pydata.org/docs/
- Statsmodels time-series tools (ADF test, decomposition)
  https://www.statsmodels.org/stable/tsa.html
- FRED data guide
  https://fred.stlouisfed.org/docs/api/fred/
- Time-series decomposition (statsmodels)
  https://www.statsmodels.org/stable/generated/statsmodels.tsa.seasonal.seasonal_decompose.html

WEEK 1 – QUANT 2: STATISTICAL MODELLING & METHODOLOGY

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Implementing and comparing ARIMA and ARIMAX models

OBJECTIVE

Implement the baseline ARIMA model and extend it to an ARIMAX model with relevant exogenous variables, comparing their performance and methodology.

KEY TASKS

1. Implement the baseline ARIMA model for the chosen economic dataset
2. Identify and incorporate relevant exogenous variables for the ARIMAX model
3. Estimate model parameters and run diagnostics (e.g. stationarity, residual analysis)
4. Compare ARIMA vs ARIMAX forecast performance using out-of-sample testing
5. Document methodology, assumptions and justification for model choices

MODEL IMPLEMENTATION

1. ARIMA MODEL (BASELINE)

- Identify appropriate (p, d, q) using ACF/PACF
- Estimate parameters and check diagnostics
- Generate out-of-sample forecasts

2. ARIMAX MODEL (EXTENSION)

- Select relevant exogenous variables (e.g. macro indicators, commodity prices, financial variables)
- Incorporate into ARIMAX model
- Estimate parameters and compare with baseline

3. MODEL EVALUATION

- Compare ARIMA vs ARIMAX using standard metrics (e.g. RMSE, MAE, MAPE)
- Use out-of-sample testing (rolling or expanding window)
- Justify model specification and final choice

USEFUL PACKAGES (PYTHON)

statsmodels	**pmdarima**	**scikit-learn**
ARIMA/ARIMAX estimation	Model selection and diagnostics	Model evaluation and metrics

EXPECTED DELIVERABLES

- Implemented ARIMA and ARIMAX models with clear methodology
- Model diagnostics and parameter estimates
- Comparison of forecast performance (ARIMA vs ARIMAX)
- Discussion of model choices and justification
- Clean, well-documented code with comments
- Written section for the final article (methodology and model comparison)

KEY REFERENCES

- Hyndman & Athanasopoulos (Forecasting) – ARIMA/ARIMAX methodology https://www.otexts.com/fpp3/
- statsmodels ARIMA/ARIMAX documentation https://www.statsmodels.org/stable/tsa.html
- pmdarima (model selection and auto_arima) https://alkaline-ml.com/pmdarima/
- scikit-learn (evaluation metrics) https://scikit-learn.org/stable/modules/model_evaluation.html

WEEK 1 – QUANT 3:
PYTHON IMPLEMENTATION & FORECAST EVALUATION

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Set up project environment and begin baseline ARIMA implementation

OBJECTIVE

Set up the Python environment, source and clean the chosen dataset, and begin implementing the baseline ARIMA model with initial exploratory analysis.

KEY TASKS

1. Set up project structure and GitHub repo with clear README
2. Source and download chosen economic dataset (e.g. FRED, BEA, Eurostat, ONS)
3. Clean and preprocess the data (handle missing values, transformations)
4. Conduct initial exploratory data analysis and visualisations (time plots, ACF/PACF)
5. Implement baseline ARIMA model with appropriate (p, d, q) selection
6. Run initial diagnostics and review results
7. Document progress in a short weekly report

KEY RESOURCES

- **statsmodels** – Time-series modelling https://www.statsmodels.org/stable/tsa.html
- **pandas** – Data handling and cleaning https://pandas.pydata.org/docs/
- **matplotlib / seaborn** – Visualisations https://matplotlib.org/ | https://seaborn.pydata.org/
- **FRED – Macroeconomic Data (US)** https://fred.stlouisfed.org/
- **BEA – US Economic Indicators** https://www.bea.gov/data
- **Eurostat – European Economic Data** https://ec.europa.eu/eurostat/
- **ONS – UK Economic Data** https://www.ons.gov.uk/

EXAMPLE ECONOMIC AREAS

Macroeconomic indicators (e.g. GDP, inflation)	Financial markets (e.g. interest rates, equity indices)	Energy & commodities (e.g. oil, gas prices)

EXPECTED DELIVERABLES

- Clean, well-structured dataset (CSV/Parquet) with clear documentation
- Exploratory data analysis notebook (including key visualisations: time plots, ACF/PACF)
- Baseline ARIMA model implemented with parameter selection and diagnostics
- Short weekly report summarising findings, any issues and next steps
- All code pushed to GitHub with clear README

USEFUL RESOURCES

- statsmodels time-series documentation https://www.statsmodels.org/stable/tsa.html
- FRED data guide https://fred.stlouisfed.org/docs/api/fred/
- Hyndman & Athanasopoulos (Forecasting) – ARIMA methodology https://otexts.com/fpp3/
- pandas data cleaning guide https://pandas.pydata.org/docs/
- matplotlib visualisation guide https://matplotlib.org/stable/plot_types/index.html

NEXT STEPS & DECISIONS

Agree first-term project, assign analysts and begin implementation

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KEY DECISIONS FOR TODAY

1. **Choose the first-term model**
   Vote on which of the three options to pursue (DNS, BSTS or ARIMA/VAR).
2. **Select economic focus area**
   Choose one of the proposed domains for the selected model (or suggest an alternative).
3. **Assign roles and responsibilities**
   Agree which analyst(s) will lead each component (data, modelling, analysis, write-up).
4. **Confirm timeline**
   Validate the six-week plan and key milestones.

FIRST-TERM GOAL

Publish one high-quality research article that replicates and modestly improves on an existing model, with clear economic insight.

IMMEDIATE NEXT STEPS

Finalise project scope	Confirm model and economic area, refine research questions.	**By 21 Oct**
Data collection and preprocessing	Gather and clean data, set up analysis environment.	**Week 1–2** (21 Oct – 3 Nov)
Model implementation	Build baseline model and initial diagnostics.	**Week 3** (4 – 10 Nov)
Results and analysis	Forecast evaluation, comparison and economic insights.	**Week 4** (11 – 17 Nov)
Article drafting	Write methodology, results and discussion sections.	**Week 5** (18 – 24 Nov)
Finalisation and publication	Incorporate feedback, prepare visuals and submit article.	**Week 6** (25 Nov – 1 Dec)

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THANK YOU

Questions & Discussion

I look forward to your thoughts and to finalising our first-term project plan.



ALIGNMENT

Confirm the chosen model and economic focus area.



ROLES

Confirm analyst responsibilities and initial tasks.



TIMELINE

Agree key milestones for the next six weeks.



NEXT STEPS

Begin Week 1 implementation and schedule check-ins.